

OmniFuse: Composite Degradation-Robust Image Fusion With Language-Driven Semantics

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Abstract—Existing image fusion methods struggle to accommodate composite degradation and do not support users flexibly modulating the semantic objects of interest. To address these challenges, this study proposes a composite degradation-robust image fusion framework with language-driven semantics, called OmniFuse. Firstly, OmniFuse establishes a novel multi-modal information fusion paradigm based on the latent diffusion model (LDM). By projecting the information fusion function into the latent space of the LDM, the information fusion process is seamlessly integrated with the diffusion process. Thus, OmniFuse fully leverages the powerful generative capabilities of LDM to eliminate composite degradation, thereby achieving highly robust image fusion. Secondly, OmniFuse develops a language-driven controllable fusion strategy to strengthen fusion flexibility. It employs a language-driven feature fusion module (LFFM) to receive the specified localization priori, dynamically aggregating multi-modal features. Within LFFM, a visual enhancement regularization is introduced to highlight objects of interest for capturing perceptual attention, while reverse semantic driving is established to strengthen their semantic attributes. Together, the visual and semantic constraints can implicitly correct the imperfect localization priori, further refining the accuracy of language-driven control. Extensive experiments demonstrate the omnipotent performance of OmniFuse, with significant advantages in robustness and flexibility compared to state-of-the-art methods.

Index Terms—Image fusion, composite degradation, latent diffusion model, language, semantics.

I. INTRODUCTION

DUE to the complexity of the real imaging environment, a single-modal image cannot provide a complete representation of the scene. This limits the ability of humans or machines to fully perceive the current environment, posing potential risks for further decision-making. In this context, multi-modal image fusion technology emerges. It aims to combine complementary

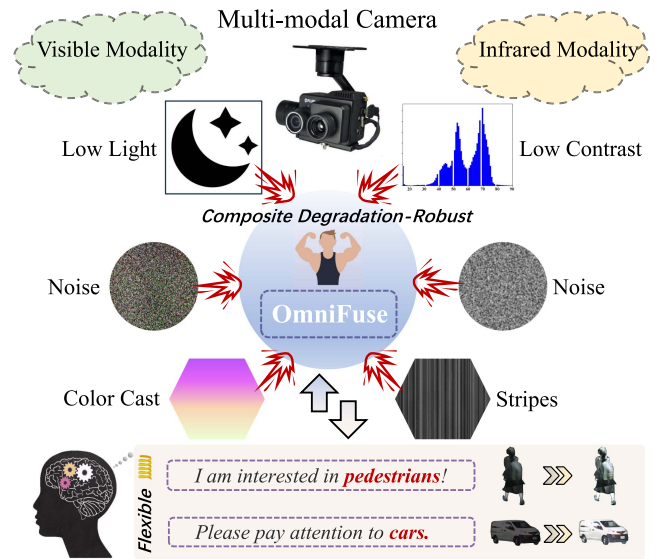


Fig. 1. The diagram of our OmniFuse. It shows omnipotent fusion performance, which can resist cross-modal composite degradation while also flexibly focusing on the semantic objects of interest according to language instructions.

information from multiple source images, generating a single fused image richer in information and stronger in representation [1]. Because of its outstanding characteristics, image fusion technology has become a key foundation of visual perception systems, being widely applied in various downstream applications [2], [3], e.g., assisted driving, intelligent security, military surveillance, etc.

In recent years, image fusion technology has made significant advancements, demonstrating excellent performance in most cases. However, two significant challenges have emerged as bottlenecks, hindering further progress. i) *Robustness barrier*. Due to hardware limitations and poor imaging conditions, real-captured multi-modal images often experience significant degradation. For example, visible images are typically affected by inadequate illumination, noise, and color cast, while infrared images suffer from low contrast, noise, and uneven stripes. These degradation factors, particularly composite ones, can interfere with the evaluation of information validity, and then affect the rationality of multi-source information retention in the fusion process. Therefore, existing fusion models often fail under unfavorable imaging environments, which contradicts the original intent of image fusion technology. ii) *Flexibility limitation*. Due to the lack of ground truth, image fusion is essentially

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The code is publicly available at <https://github.com/HaoZhang1018/OmniFuse>.

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a no-reference perception problem. This means that different humans and machines have varying requirements for the fused images. However, current image fusion is usually solved within closed fusion frameworks, where the fusion function becomes fixed once the optimization is completed. Such closed fusion frameworks lack flexibility and cannot meet the personalized visual and semantic perception needs of different users, whether human or machine. Next, we first outline the shortcomings of existing methods in these two aspects and then propose our contributions.

A. Previous Arts

The degradation in actual captured infrared and visible images poses a significant robustness challenge for existing multi-modal image fusion methods, which has long hindered their practical application. Recently, some studies have targeted degradation-robust fusion. For instance, DeRUN [4] employs ℓ_1 -norm regularization on the gradient term to reduce noise degradation, while DIVFusion [5] addresses low-light degradation by separating inappropriate illumination. Similarly, our preliminary version DDBF [6] addresses low-light degradation by utilizing infrared images to guide the visible information restoration, synergizing low-light enhancement with information fusion. However, these methods focus on single-type degradation, limiting their applicability to complex real-world scenarios with multiple types of degradation. To delve deeper, Text-IF [7] explores multi-type degradation by using text as a switch to selectively activate restoration functions, yet it can only address one degradation type per run. MRFS [8] adopts simple data augmentation strategies (e.g., gamma transformation, contrast stretching) to pre-enhance source images, and then regress to correct illumination and remove noise during fusion. However, due to the roughness of data augmentation, the fused results often contain artifacts. VIFNet [9] proposes an inconsistency fusion strategy to combine infrared and visible information, aiming to remove haze and correct low contrast through supervised regression with clean image guidance. Notably, such hard regression is very difficult and may lead to blurring of the fused results. All the aforementioned fusion methods assume that only one of the paired infrared and visible images is degraded, thereby allowing the clean image to supplement the degraded one. Unfortunately, this ideal assumption rarely holds in practice, still posing challenges to their robustness. Overall, current research on fusion robustness is still in a relatively preliminary stage. *Given practical scenarios where both multi-source images exhibit composite degradation, exploring more robust fusion models is highly desirable.*

Flexibility limitation is another crucial factor impacting the practical deployment of fusion models. Given that different users may have distinct visual and semantic preferences, closed fusion frameworks are inadequate for addressing the dynamic requirements of real-world applications. Recently, some methods have begun exploring controllable fusion mechanisms. Our preliminary DDBF [6] uses conditional generative adversarial networks (CGANs) for controllable image fusion. It enables varying information restoration through adjustable enhancement ratios, further deriving diverse fusion appearances. Another

example, Text-IF [7], leverages conditional text prompts based on contrastive language-image pre-training (CLIP) [10] to activate different information restoration functions, achieving customizable fusion outcomes. Differently, TC-MoA [11] enhances model flexibility through multi-expert prompt learning, where each expert serves as a tuning adapter to generate task-specific prompts, fulfilling a range of fusion tasks. However, these controllable methods primarily focus on global visual control, lacking fine-grained semantic awareness. For instance, they can not dynamically modulate semantic objects, e.g., pedestrians and vehicles, which are precisely what is more needed in real-world applications. Therefore, *developing a controllable fusion model with semantically fine-grained perception ability is both practical and necessary.*

B. Our Contributions

To break the robustness barrier and flexibility limitation in image fusion, we propose a composite degradation-robust image fusion framework with language-driven semantics, named OmniFuse. Firstly, we construct a novel latent diffusion model (LDM)-based information fusion paradigm to enhance the robustness of the fusion model. Specifically, we project the information fusion process to the latent space of the LDM, and seamlessly integrate the information diffusion and fusion. In it, the LDM is allowed to pre-learn the removal of composite degradation under conditional guidance and integrate this priori knowledge into the cross-modal latent diffusion network. Then, the fusion module can benefit from clean features in the latent space, focusing on performing higher-quality multi-modal information aggregation. This innovative coupling strategy between LDM and information fusion effectively avoids the dilemma of not being able to retrain the LDM due to the lack of ground truth in the field of image fusion. As a result, the fusion model is driven to be highly robust to the real scenario where both multi-source images exhibit composite degradation. Secondly, to strengthen flexibility, we develop a language-driven controllable fusion strategy, transforming OmniFuse into an open fusion framework. Concretely, a language-driven feature fusion module (LFFM) is designed to receive the location priori from the vision-language large model (vLLM) [12]. It can dynamically aggregate multi-modal features based on language instructions, achieving preference modulation for specific objects. Within LFFM, a customized visual enhancement regularization is constructed to highlight objects of interest through frequency separation-based selective enhancement. Meanwhile, a semantic segmentation model is introduced after the fused image to guide a semantic injection during the fusion process. Through collaborative optimization, the imperfect location priori can be reversely corrected, thereby accurately satisfying the needs of downstream decision-making tasks to focus on specific semantic objects. Overall, OmniFuse exhibits robustness in harsh environments while flexibly meeting users' various visual and semantic needs, showing comprehensive fusion performance beyond that of current methods, as illustrated in Fig. 1.

To conclude, the contributions of this work include the following four aspects:

- 1) We propose a composite degradation-robust image fusion framework with language-driven semantics, offering an integrated solution to overcome robustness barriers and flexibility limitations in current fusion methods.
- 2) We construct a novel multi-modal information fusion paradigm based on the LDM, seamlessly integrating information fusion with latent diffusion to achieve robust image fusion in extreme composite degradation scenarios. To our knowledge, this is the first collaborative realization of composite degradation removal and multi-modal information fusion under the LDM framework.
- 3) A language-drive controllable fusion strategy is developed, enabling users to flexibly control the fusion process through language instructions. Through the guarantee of both visual and semantic aspects, it successfully achieves a semantically fine-grained perception ability that no existing method can match.
- 4) We conduct extensive fusion comparisons and language control validation across multiple scenarios. Qualitative and quantitative results demonstrate that our OmniFuse can effectively defend against various composite degradation, and flexibly offer promising visual and semantic gains for the objects of interest.

A preliminary version of this paper is DDBF [6]. The new contributions are mainly from two aspects. First, the range of degradation types that the fusion model can adapt to is significantly expanded. Specifically, the preliminary version, DDBF, mainly focuses on the information degradation caused by inadequate illumination in visible modality. In contrast, the current version, OmniFuse, can not only address the composite degradation of various types contained in the visible modality but also handle extremely harsh environments where both infrared and visible modalities suffer from degradation. For instance, OmniFuse can adapt to inadequate illumination, noise, and color cast in visible images, and low contrast, noise, and uneven stripes in infrared images. Thus, compared to DDBF, OmniFuse demonstrates greater application reliability in real-world scenarios. Second, OmniFuse also achieves an effective improvement in fusion flexibility compared to the preliminary version. Concretely, DDBF utilizes discrete numbers as control conditions to enable global adjustment of the illumination level. By comparison, OmniFuse incorporates language instructions to achieve a more nuanced and open dynamic information fusion. Besides, by applying constraints in both visual and semantic aspects, OmniFuse can shift dynamic control from the global to the local level, achieving promising semantically fine-grained perception.

II. RELATED WORK

A. Diffusion-Based Fusion

Diffusion models have demonstrated strong generative capabilities through progressive distribution estimation, significantly improving low-level vision tasks such as image synthesis [13] and restoration [14]. Most recently, diffusion models have also been introduced into the image fusion task to enhance multi-modal information [15] extraction and aggregation. Both

Dif-fusion [16] and DCAFuse [17] use the diffusion model as the feature extractor, combining it with the convolutional neural network to achieve aggregation of multi-modal global and local features. However, such a strategy makes very limited use of the generative capability of the diffusion model, resulting in sub-optimal fusion performance. Subsequently, DDFM [18] treats infrared and visible images as generative conditions. By leveraging expectation-maximization during the inference phase of the denoising diffusion probabilistic model, driving the preservation and fusion of valuable multi-modal information. FusionDiff [15] utilizes multi-focus images as guiding information to iteratively denoise the initial random noise, ultimately producing the full clear fused image. Diff-IF [19] adopts an alternative strategy that selects high-quality fused results from existing methods as training targets, embedding fusion priori into the diffusion process for improved performance. However, all these methods execute diffusion in image space, which potentially leads to inadequate intrinsic characteristic estimation. LFDT-Fusion [20] addresses this by shifting fusion to the latent space through the LDM. It employs a fusion head to aggregate latent features from different layers of the denoising network, enhancing the completeness of feature extraction and fusion. Notably, current diffusion-based fusion methods do not consider the degradation contained in multi-modal images when using the image generation priori of diffusion models. This makes it difficult for these methods to cope with real-world scenarios that contain cross-modal composite degradation. By comparison, our OmniFuse goes beyond the image generation priori by embedding the information restoration priori into the latent diffusion process, enabling more robust fusion. This enables it to remain reliable even in harsh environments with composite degradation.

B. Language-Driven Fusion

Language, a crucial vehicle of human cognition, is regarded as one of the most important forms of interaction for intelligent systems. With the rapid development of vLLM technology, language instructions have been integrated into various computer vision tasks, enabling flexibly controllable image synthesis [21] and semantic decision-making [22], etc. In the task of image fusion, researchers have also begun exploring the embedding of language to build open fusion frameworks. For example, FILM [23] and FTP [24] pre-generate textual descriptions of scenes, which are then encoded and fused with multi-modal image features. In these works, language serves as auxiliary information to enhance scene representation but does not modulate the fusion process. LDL [25] uses CLIP [10] to convert language into an optimization regularization, improving the rationality of fusion by ensuring the fused image aligns with predefined textual descriptions. Similarly, PromptFusion [26] also leverages language for fusion constraints, but unlike LDL, it simultaneously uses both positive and negative prompts to construct contrastive learning, significantly enhancing the feature representation of the fused image. MTG-Fusion [27] integrates the advantages of the above methods, using language as both a complement for multi-modal image aggregation and a guiding constraint for fusion optimization, achieving improved performance.

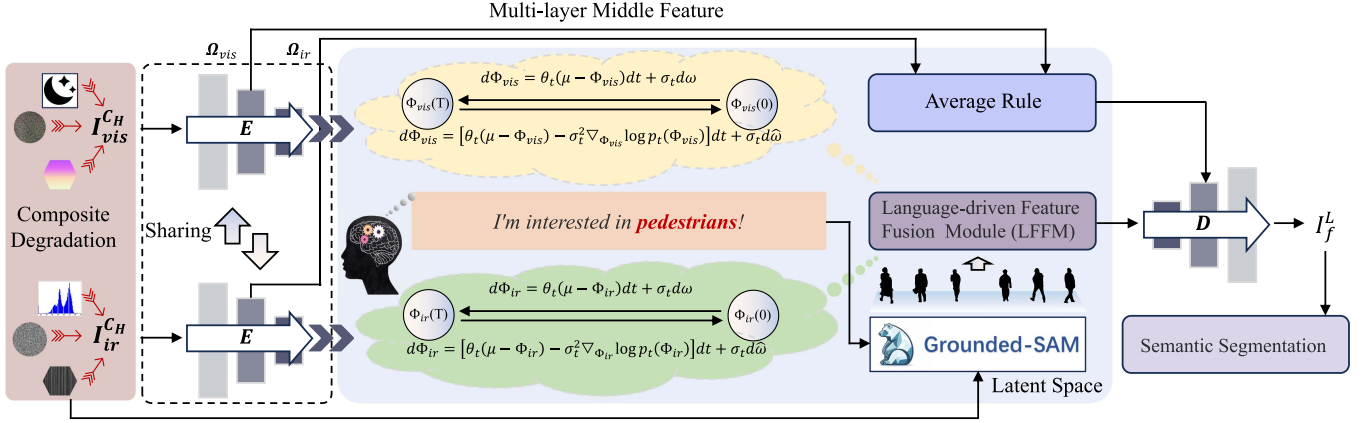


Fig. 2. Overall framework of the proposed OmniFuse for composite degradation-robust image fusion with language-driven semantics.

TABLE I
LIST OF NOTATIONS IN THIS PAPER

Notations	Paraphrase
LDF	Latent diffusion fusion
$LDCF$	Language-driven controllable fusion
$LFFM$	Language-driven feature fusion module
Seg	Segmentation network
I_M^{CH}	M-modal image with H -type degradation
L	Language instructions
E, D	Encoder, decoder
$\Phi_M^{CH}, \Omega_M^{CH}$	Latent features, multi-layer middle features
$\Phi_M^{CH \rightarrow 0}$	Diffusion purified M-modal latent features
$\tilde{I}_M^{CH \rightarrow 0}$	Diffusion purified M-modal image
Φ_f	Fused latent features
Ω_f	Fused multi-layer middle features
I_f	Fused image
Φ_f^L	Fused latent features modulated by language instructions
I_f^L	Fused image modulated by language instructions
$\tilde{I}_{f,\zeta}^L$	frequency separation-based visual reference
$\mathcal{Q}_L$	language-specified location priori
$\zeta, \bar{\zeta}$	Labels of language-specified region and other regions
$\mathcal{N}_{high}, \mathcal{N}_{low}$	High-frequency and low-frequency components
$\hat{Y}, \bar{Y}$	Segmentation output, label

However, LDL, PromptFusion, and MTG-Fusion still lack the ability for flexible control, as language only operates within the optimization space. Furthermore, Text-IF [7] uses language prompts to activate various information restoration abilities, but its dynamic regulation focuses on global appearance changes rather than fine-grained semantic control. To improve the modulation fineness, our OmniFuse incorporates the state-of-the-art Grounded-SAM [12] to mine the fine-grained localization priori for language-specified objects. Building on this, our OmniFuse further implicitly corrects the imperfect location priori, thereby accurately enhancing the object of interest in both visual and semantic aspects.

III. METHODOLOGY

Our OmniFuse can enhance the robustness of the fusion model against composite degradation and achieve language-driven fine-grained control of the semantic objects of interest. In this section, we provide the problem formulation and introduce the

TABLE II
DEGRADATION TYPE STATISTICS IN CURRENT INFRARED AND VISIBLE IMAGE FUSION DATASETS [28], [29], [30], [31]

Dataset	Visible Modality			Infrared Modality		
	Low	Light	Noise	Color	Cast	Stripes
MFNet	✓	✓	✓	✓	✓	✓
LLVIP	✓	✓	✓	✓	✓	✓
FMB	✓	✓	✓	✓	✓	✓
RoadScene	✓	✓	✓	✓	✓	✓

latent diffusion-based fusion paradigm and the language-driven controllable fusion strategy in detail. Notably, since this paper involves relatively many variables, we provide a list of notations for querying, as demonstrated in Table I.

A. Problem Formulation

Focusing on addressing the robustness barrier and flexibility limitation, we propose the OmniFuse for composite degradation-robust image fusion with language-driven semantics, as shown in Fig. 2. Now, let us formally define this research problem. The multi-modal images captured in harsh environments are designated as $\{I_{vis}^{CH}, I_{ir}^{CH}\}$, where vis and ir represent infrared and visible modalities. CH denotes the composite degradation, which is specified as inadequate illumination, noise, and color cast in the visible modality, and as low contrast, noise, and uneven stripes in the infrared modality. As shown in Table II, they essentially cover the common degradation types in existing infrared and visible image fusion datasets, and can be considered complete to some extent. In particular, $H = 0$ indicates the absence of degradation. To enhance the robustness of the fusion model against the composite degradation, we first constructed a new multi-modal information fusion paradigm based on the LDM. The core of this paradigm lies in the coupling of the diffusion process and the fusion process so that the aggregation of multi-modal information can benefit from the information restoration priori. Thus, we initially utilize the LDM to perform diffusion purification of $\{I_{vis}^{CH}, I_{ir}^{CH}\}$, embedding the information restoration priori into the latent space. Then, LFFM is introduced into the latent space to aggregate the purified multi-modal features, and use the decoder to remap the fused features back

to the image space. We refer to this process as latent diffusion fusion (LDF), which can produce a clean fused image free from composite degradation $I_f = LDF(I_{vis}^{C_H}, I_{ir}^{C_H})$.

Notably, if OmniFuse relies solely on LDF, it remains a closed fusion framework that cannot meet users' varying preferences. Therefore, to strengthen the flexibility of our method, we further propose a language-driven controllable fusion (LDCF) strategy. It uses language instructions to drive LFFM, enabling spatial dynamic aggregation of multi-modal information. Specifically, LFFM provides a conditional control interface that receives the fine-grained localization priori from Grounded-SAM. Based on this priori, LFFM can flexibly highlight the semantic objects of interest according to the user's language instructions. Therefore, the controllable fusion process of our proposed method can be further formalized as $I_f^L = LDCF(I_{vis}^{C_H}, I_{ir}^{C_H}, L)$, where L denotes the language instructions. To ensure the quality of language control, we establish modulation constraints from both visual and semantic aspects. Firstly, the visual enhancement regularization is constructed based on frequency separation, highlighting the language-specified objects of interest and satisfying users' visual perception needs. Meanwhile, semantic reverse guidance is established through the driving of a segmentation model, reinforcing the semantics of the objects of interest. It is worth emphasizing that we use segmentation labels to locate language-specific objects when constructing visual and semantic constraints, which implicitly corrects the imperfect fine-grained localization priori provided by Grounded-SAM. Ultimately, the robustness and flexibility of OmniFuse are well assured, allowing it to be reliably applied in adverse real-world environments and to meet different visual and semantic preferences of both humans and machines.

B. Latent Diffusion-Based Fusion Paradigm

1) *Information Restoration Priori Embedding*: Embedding the information restoration priori into the LDM is a prerequisite for achieving composite degradation-robust fusion. As shown in Fig. 2, the information restoration is achieved through the diffusion process in the latent space. Therefore, it is necessary to first pre-train an autoencoder to establish a high-quality latent space. Formally, the encoder E maps the multi-modal images $I_M^{C_H}$ to produce the latent features $\Phi_M^{C_H}$, where M indicates the modality type. Meanwhile, to minimize the damage to scene information during the mapping processes, we also output the multi-layer middle features $\Omega_M^{C_H}$ by skip connection, which refer to the collection of features from each layer of the encoder. Therefore, the mapping process of the encoder can be formalized as:

$$\Phi_M^{C_H}, \Omega_M^{C_H} = E(I_M^{C_H}). \quad (1)$$

Then, the decoder receives the latent features and the multi-layer middle features, thereby mapping them back from the latent space to the image space:

$$\hat{I}_M^{C_H} = D(\Phi_M^{C_H}, \Omega_M^{C_H}). \quad (2)$$

Unlike conventional self-reconstruction optimization, our OmniFuse proposes two specific requirements for the encoding

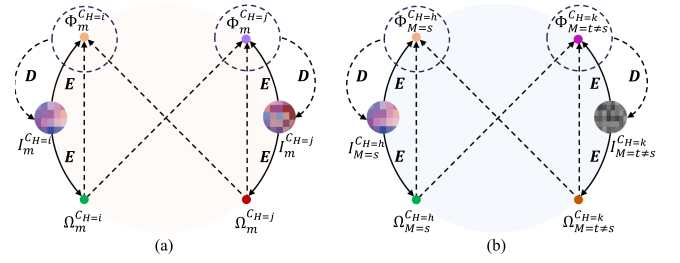


Fig. 3. Illustration of the cross-modal feature reorganization mechanism, in which the mosaic patterns denote degraded images. (a) The same-modal reorganization constraint; (b) the cross-modal reorganization constraint.

and decoding process. First, the degradation attribute should be concentrated as much as possible in the latent features. Specifically, the multi-layer middle features are quite massive, so applying diffusion to remove degradation from them would be highly resource-intensive. In this context, if the degradation attribute is transferred to the latent features, we can apply diffusion only to the small-scale latent features, making the process highly efficient. Second, the multi-layer middle features should focus on supplementing scene information and must exhibit cross-modal invariance during decoding. Concretely, the role of the multi-layer middle features is to prevent the loss of scene information during the encoding processes. Thus, by ensuring the cross-modal invariance, we are permitted to use a simple averaging rule to fuse the multi-modal middle features, so that LFFM can focus on the dynamic aggregation of the more critical latent features.

To meet these two requirements, we propose a cross-modal feature reorganization mechanism to guide the optimization of the autoencoder, as shown in Fig. 3. First, for same-scene and same-modality images $I_M^{C_H}$ with different degradation, exchanging the multi-layer middle features should not affect the image reconstruction. The corresponding same-modal reorganization loss $\mathcal{L}_{SMR}$ can be formalized as:

$$\mathcal{L}_{SMR} = \sum_m \sum_i \sum_j \left\| I_m^{C_H=i} - D(\Phi_m^{C_H=i}, \Omega_m^{C_H=j}) \right\|, \quad (3)$$

here m indicates the modality type, which specifically refers to visible and infrared modalities in this work. i and j denote the index of degradation types, where $i = j = 0$ indicates no degradation. Clearly, this constraint establishes a strong association between the degradation attribute and the latent features. In other words, regardless of the degradation type in the image from which the multi-layer middle features originate, the reconstructed results depend solely on the source of the latent features. Second, to meet the needs of subsequent fusion, we further construct a cross-modal reorganization loss $\mathcal{L}_{CMR}$ to constrain the cross-modal invariance of the multi-layer middle features during decoding:

$$\mathcal{L}_{CMR} = \sum_s \sum_t \sum_h \sum_k \left\| I_{M=s}^{C_H=h} - D(\Phi_{M=s}^{C_H=h}, \Omega_{M=t \neq s}^{C_H=k}) \right\|, \quad (4)$$

where h and k indicate the degradation type, s and t are the index of modality types. Intuitively, for cross-modal images of

the same scene, this constraint ensures that the reconstructed results are independent of the modality from which the multi-layer middle features originate. In summary, the total loss used for training the autoencoder is:

$$\mathcal{L}_{AE} = \mathcal{L}_{SMR} + \mathcal{L}_{CMR}. \quad (5)$$

By optimizing $\mathcal{L}_{AE}$, the transfer of degradation attributes is effectively promoted, and the modality-invariant property of the multi-layer middle features during decoding is ensured.

Next, we implement diffusion optimization in the latent space based on the trained autoencoder, to drive the embedding of the information restoration priori. Note that since the degradation attribute has been primarily encoded into the latent features, the diffusion optimization only needs to be applied to latent features Φ_M^{CH} . Specifically, we adopt the latent diffusion framework based on the mean-reverting stochastic differential equations, as proposed in IR-SDE [32] and Refusion [33]. The forward process is formalized as:

$$d\Phi_M = \theta_t(\mu - \Phi_M)dt + \sigma_t d\omega, \quad (6)$$

where dt represents the time step, $d\omega$ indicates the Gaussian noise. θ_t denotes the parameter that controls the mean-reversion speed, and σ_t represents the dispersion coefficient that controls the noise addition degree. μ is the mean value, ensuring that after a certain number of time steps, the forward process will converge to a stable Gaussian distribution with a mean value of μ . Notably, here we simplify Φ_M^{CH} to Φ_M for more conveniently representing variables in the diffusion process. Corresponding to the forward process, the reverse-time stochastic differential equation can be formalized as:

$$d\Phi_M = [\theta_t(\mu - \Phi_M) - \sigma_t^2 \nabla_{\Phi_M} \log p_t(\Phi_M)]dt + \sigma_t d\hat{\omega}, \quad (7)$$

in which $\nabla_{\Phi_M} \log p_t(\Phi_M)$ denotes the score function. To perform reverse-time inference, solving the unknown $\nabla_{\Phi_M} \log p_t(\Phi_M)$ is indispensable. IR-SDE [32] proves that the marginal distribution $p_t(\Phi_M)$ can be calculated from Eq. (7) by setting $\sigma_t^2/\theta_t = 2\lambda^2$:

$$\begin{cases} p_t(\Phi_M) = \mathcal{N}(\Phi_M(t)|m_t, v_t), \\ m_t(\Phi_M) := \mu + (\Phi_M(0) - \mu)e^{-\bar{\theta}_t}, \\ v_t := \lambda^2(1 - e^{-2\bar{\theta}_t}), \end{cases} \quad (8)$$

where $\bar{\theta}_t = \int_0^t \theta_z dz$. Therefore, the conditional score $\nabla_{\Phi_M} \log p_t(\Phi_M)$ can be obtained:

$$\nabla_{\Phi_M} \log p_t(\Phi_M|\Phi_M(0)) = -\frac{\Phi_M(t) - m_t(\Phi_M)}{v_t}. \quad (9)$$

During training, $\Phi_M(0)$ can be considered as an available high-quality latent feature $\Phi_M^{CH=0}$. By reparameterization, $\Phi_M(t)$ can be represented as $\Phi_M(t) = m_t(\Phi_M) + \sqrt{v_t}\epsilon_t$, where ϵ_t is the standard Gaussian noise. Thus, the conditional score can be further obtained by $\nabla_{\Phi_M} \log p_t(\Phi_M|\Phi_M(0)) = -\epsilon_t/\sqrt{v_t}$. Obviously, we can train a denoising network with the paired $[\Phi_M^{CH=0}, \Phi_M^{CH \neq 0}]$ to estimate the noise, and finally achieve complete reverse-time inference. More importantly, this allows us to avoid considering the specific degradation process. As long as there are available datasets with paired clean-degraded images,

the degradation listed in Table II can all be removed. Now, the information restoration priori has been embedded into the latent space, creating the conditions necessary for further implementing latent diffusion fusion.

2) *Latent Diffusion Fusion*: In our proposed fusion paradigm, the restored feature $\Phi_M(0)$ obtained through the complete reverse-time inference process is specified as multi-modal latent features $\Phi_{vis}(0)$ and $\Phi_{ir}(0)$. They can be approximately considered as clean:

$$\begin{cases} \Phi_{vis}(0) \rightarrow \Phi_{vis}^{CH=0}, \\ \Phi_{ir}(0) \rightarrow \Phi_{ir}^{CH=0}. \end{cases} \quad (10)$$

As shown in Fig. 2, we propose the LFFM to aggregate these multi-modal latent features, which can be formalized as:

$$\Phi_f = LFFM(\Phi_{vis}(0), \Phi_{ir}(0)). \quad (11)$$

Notably, to reduce the solution space, LFFM opts to generate two weight matrices, W_{vis} and W_{ir} , for fusion rather than directly outputting the fused features. Thus, Eq. (11) can be further rewritten as:

$$\Phi_f = W_{vis} \odot \Phi_{vis}(0) + W_{ir} \odot \Phi_{ir}(0). \quad (12)$$

Moreover, with the cross-modal reorganization loss in Eq. (4) ensuring the cross-modal invariance of multi-layer middle features during decoding, a simple averaging rule is adopted for fusing them:

$$\Omega_f = (\Omega_{vis}^{CH} + \Omega_{ir}^{CH})/2. \quad (13)$$

With all the necessary variables prepared, the decoder can now be used to reconstruct the fused image:

$$I_f = D(\Phi_f, \Omega_f). \quad (14)$$

Next, we discuss the design of the fusion-related loss functions, to guarantee effective optimization of multi-modal information fusion. In the field of multi-modal image fusion, since the ground truth does not exist, current methods usually preserve valuable information by reducing the similarity distance between the fused image and the multi-source images. However, when the source images contain degradation, such a constraint will inevitably lead to the transmission of degradation into the fused image as well. In this dilemma, the ingenuity of our proposed latent diffusion-based fusion paradigm is that it can provide clean source guidance for fusion optimization. Specifically, we can utilize the trained autoencoder and diffusion model to generate a clean version of the source images:

$$\begin{cases} \hat{I}_{vis}^{CH \rightarrow 0} = D(\Phi_{vis}(0), \Omega_{vis}^{CH}), \\ \hat{I}_{ir}^{CH \rightarrow 0} = D(\Phi_{ir}(0), \Omega_{ir}^{CH}). \end{cases} \quad (15)$$

With $\hat{I}_{vis}^{CH \rightarrow 0}$ and $\hat{I}_{ir}^{CH \rightarrow 0}$, we can begin designing the fusion-related loss functions. In line with our preliminary version [6], we first adopt the idea of explicit information preservation to propose a significant contrast loss $\mathcal{L}_{SC}$ and a sharpened texture loss $\mathcal{L}_{ST}$. The significant contrast loss $\mathcal{L}_{SC}$ is defined as:

$$\mathcal{L}_{SC} = \left\| I_f - \max(\hat{I}_{vis}^{CH \rightarrow 0}, \hat{I}_{ir}^{CH \rightarrow 0}) \right\|, \quad (16)$$

where $\max$ denotes the maximum value function, which signifies selecting the most salient pixel intensity from the multi-modal images as the target for approximation. $\mathcal{L}_{SC}$ is particularly important for preserving high-salient objects in the imaging scene, e.g., pedestrians and vehicles with thermal radiation. Similarly, the sharpened texture loss $\mathcal{L}_{ST}$ is formalized as:

$$\mathcal{L}_{ST} = \left\| \nabla I_f - \max(\nabla \hat{I}_{vis}^{C_{H \rightarrow 0}}, \nabla \hat{I}_{ir}^{C_{H \rightarrow 0}}) \right\|. \quad (17)$$

Intuitively, $\mathcal{L}_{ST}$ focuses on selecting and preserving the richest texture structures at the pixel level. It complements $\mathcal{L}_{SC}$, thereby sufficiently preserving high-value scene information from both the brightness distribution and the structural attribute of the image. Moreover, we constrain the color properties of the fused image to ensure that its essential hue is consistent with the visible modality for visual perception. The cosine similarity-based hue fidelity loss $\mathcal{L}_{HF}$ is defined as:

$$\mathcal{L}_{HF} = 1 - \sum_a \sum_b \frac{\sum_c I_{f,a,b,c} \times \hat{I}_{vis,a,b,c}^{C_{H \rightarrow 0}}}{\sum_c \sqrt{I_{f,a,b,c}^2} \times \sqrt{\hat{I}_{vis,a,b,c}^{C_{H \rightarrow 0}^2}}}, \quad (18)$$

in which a , b , and c represent the pixels in the a -th row, b -th column, and c -th channel, respectively. $\mathcal{L}_{HF}$ eliminates the dimensional differences through normalization along the channel dimension, allowing the color constraint to focus on the intrinsic hue, thereby effectively ensuring the color fidelity of the fused image. The total loss for LDF is summarized as:

$$\mathcal{L}_{LDF} = \mathcal{L}_{SC} + \mathcal{L}_{ST} + \mathcal{L}_{HF}. \quad (19)$$

So far, our OmniFuse has been capable of achieving highly robust multi-modal image fusion, not only defending against various composite degradation but also delivering a high-quality fusion appearance.

C. Language-Driven Controllable Fusion Strategy

1) *Fine-Grained Location Priori Injection*: How to transition the fusion framework from a closed-form to an open-form one is key to meeting the requirements of various applications. To this end, we introduce language instructions to drive the LFFM, enabling it to control the fusion process of multi-modal features based on specific requirements. As shown in Fig. 2, the goal is to focus more on the language-specified objects of interest in the imaging scene, making them exhibit more prominent visual performance and semantic attributes. However, direct language encoding embedding implies that the LFFM must be capable of simultaneously performing language understanding, object localization, and fusion control, which presents a considerable challenge. Fortunately, Grounded-SAM has already built a bridge between vision and language, which integrates Grounding DINO [34] and Segment Anything Model [35] for zero-shot object localization via language prompts. Naturally, it can provide the fine-grained location priori as an effective initialization clue to help OmniFuse locate the region of interest and enhance its visual representation. Therefore, our LFFM is designed as a conditional generative network with the condition of the language-specified location priori ϱ_L , as shown in Fig. 4.

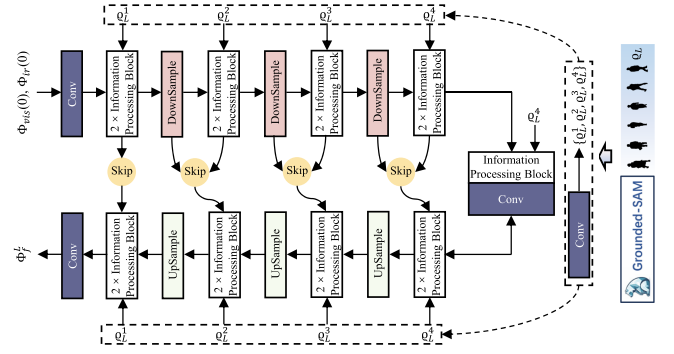


Fig. 4. Structural diagram of the language-driven feature fusion module. We use the Grounded-SAM [12] to generate fine-grained location priori.

Specifically, LFFM modifies the four-scale UNet architecture used in Stable Diffusion [21]. At each scale, there is an information processing block composed of the ResNet and Transformer working in tandem. Between different scales, downsampling and upsampling layers are adopted to change the feature size. To align with the feature size of UNet, we employ continuous convolution to obtain the location priori at different scales: $\{\varrho_L^1, \varrho_L^2, \varrho_L^3, \varrho_L^4\} = \text{Conv}(\varrho_L)$. Subsequently, the multi-scale priori are densely integrated into each information processing block, to ensure adequate conditional coupling. Therefore, Eq. (11) can be extended to:

$$\Phi_f^L = LFFM(\Phi_{vis}(0), \Phi_{ir}(0), \varrho_L), \quad (20)$$

where Φ_f^L denotes the fused latent features modulated by language instructions. Finally, the language-driven fused images can be obtained according to:

$$I_f^L = D(\Phi_f^L, \Omega_f). \quad (21)$$

2) *Visual Enhancement Regularization*: Recall the flexibility goal of OmniFuse, which is to enhance the visual salience of the object of interest based on language instructions. To align with this desired control function, we need to impose regularization on the visual appearance. Formally, we designate the region occupied by the language-specified semantic object as ζ . First, I_f^L should globally inherit the desirable properties of I_f^* that is obtained by solving Eq. (19), to defend against cross-modal composite degradation. Simultaneously, the saliency of region ζ should be enhanced to capture the user's visual attention. Clearly, the global robustness assurance and local saliency enhancement impose overlapping constraints on region ζ , which may lead to optimization conflicts. To address this challenge, we define a frequency separation-based visual reference:

$$\hat{I}_{f,\zeta}^L = \aleph_{low}(I_f^*) + \eta \cdot \aleph_{high}(I_f^*)_{\zeta} + \aleph_{high}(I_f^*)_{\bar{\zeta}}, \quad (22)$$

where $\aleph_{low}$ and $\aleph_{high}$ represent the high-frequency and low-frequency components, respectively. ζ and $\bar{\zeta}$ denote the language-specified region and other regions, and η is the gain factor. Intuitively, the global consistency of the low-frequency components provides fundamental assurance for attributes such as brightness and color in the fused image. Then, we tackle the region of interest ζ and the other regions $\bar{\zeta}$ separately in the

high-frequency components, to drive contrastive visual enhancement. Specifically, for the region of non-interest, we preserve its high-frequency components unchanged. In contrast, for the region of interest, we enhance its high-frequency components using a gain factor η . This strategy not only enables the objects in that region to present sharper texture details but also achieves greater saliency through residual injection. In this work, η is set to 10 to achieve relatively good performance. Therefore, the visual enhancement regularization can be formalized as:

$$\mathcal{L}_{\text{VER}} = \left\| I_f^L - \hat{I}_{f,\zeta}^L \right\|. \quad (23)$$

Notably, ζ is specified as the segmentation label indicating the location of language-specified objects. It is distinct from the location priori provided by Grounded-SAM in Fig. 4. The motivation for this choice lies in the fact that Grounded-SAM is an open-set model, and its provided location priori may not be perfect. In this context, using the segmentation labels to construct the visual enhancement regularization can implicitly correct the imperfect location priori.

3) *Semantic Reverse Guidance*: While applying the visual enhancement regularization can highlight the appearance saliency of the region of interest, it cannot guarantee their semantic attributes. As a result, it potentially undermines the fused image's suitability for machine decision-making applications. To address this issue, OmniFuse further introduces a semantic reverse guidance mechanism to improve the semantic quality of the specified objects in alignment with language instructions. Concretely, an intensive decision task, semantic segmentation, is employed to achieve the imaging scene understanding based on I_f^L :

$$\hat{Y} = \text{Seg}^*(I_f^L), \quad (24)$$

where $\hat{Y}$ is the predicted classification probability, and Seg^* is the frozen segmentation network that is specified as Seg-Next [36]. Obviously, the modulated fused image as input will directly affect the accuracy of semantic segmentation. Therefore, we can reversely guide the modulation of the fused image by constraining the segmentation scores. To make semantic modulation conform to language instructions, we establish the semantic guidance loss based on the weighted cross entropy:

$$\mathcal{L}_{\text{SG}} = - \sum_{c=1}^C w_c \cdot Y_c \cdot \log \hat{Y}_c, \quad (25)$$

in which C denotes the number of semantic categories, and Y refers to the segmentation label. w_c is the category-related weighting coefficient, which is associated with language instructions. Specifically, the semantic category specified by the language instructions should be assigned a higher weight to enhance its priority during optimization, thereby focusing on guiding the semantic attribute injection into the region ζ . Formally, w_c is defined as:

$$w_c = \begin{cases} \alpha \cdot w^{ib}, & c \leftrightarrow L, \\ w^{ib}, & \text{else}, \end{cases} \quad (26)$$

where α denotes the gain factor, and $c \leftrightarrow L$ is a matcher used to indicate that the current category c matches the language

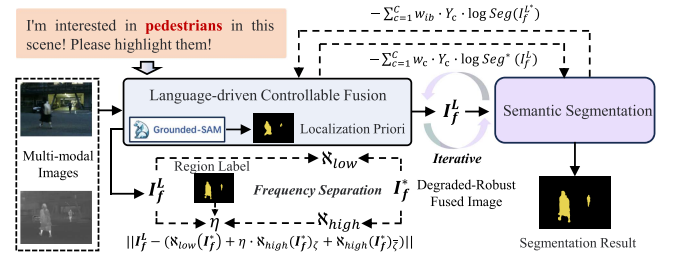


Fig. 5. The cooperative optimization between fusion and segmentation.

instruction L . w^{ib} is the initial category correction coefficient, which is commonly employed to overcome the imbalance in segmentation samples [37]. Intuitively, if the current category matches the language, the gain factor α is used to amplify the corresponding coefficient, thereby strengthening the semantic guidance for this category. In this work, α is set to 10.

However, a fixed segmentation network cannot adapt to the continuously updated I_f^L , hindering the ongoing enhancement of semantic guidance. To address this challenge, we further establish a cooperative optimization mechanism between fusion and segmentation. On the one hand, the language-driven controllable fusion loss can be summarized as:

$$\begin{aligned} \mathcal{L}_{\text{LDCF}} &= \mathcal{L}_{\text{VER}} + \mathcal{L}_{\text{SG}} \\ &= \left\| I_f^L - \hat{I}_{f,\zeta}^L \right\| - \sum_{c=1}^C w_c \cdot \text{Seg}^*(I_f^L)_c \cdot \log \hat{Y}_c. \end{aligned} \quad (27)$$

At this point, the segmentation network is frozen. On the other hand, we establish a semantic segmentation loss to drive the optimization of the segmentation network:

$$\mathcal{L}_{\text{SS}} = - \sum_{c=1}^C w_{ib} \cdot Y_c \cdot \log \text{Seg}(I_f^{L*}), \quad (28)$$

where I_f^{L*} represents the fused image generated by the fixed language-driven controllable fusion process. Clearly, Eqs. (27) and (28) are coupled, which allows us to achieve the mutual reinforcement between fusion and segmentation in an iterative manner, as shown in Fig. 5. Such an iterative optimization significantly enhances the semantic attributes of the language-specified object. At the same time, it also helps to correct the imperfection in the location priori.

It is worth emphasizing that our OmniFuse addresses the challenges of robustness and flexibility in an integrated manner rather than separately. In other words, LDF and LDCF are not independent. $\mathcal{L}_{\text{LDCF}}$ in Eq. (27) not only inherits the robustness assurance provided by $\mathcal{L}_{\text{LDF}}$ in Eq. (19) but also promotes the flexibility of image fusion according to language instructions. Under the configuration of the initialization inheritance of network parameters, the optimization of LDF and LDCF can be represented as shown in Fig. 6. It can be seen that their function surfaces are similar overall, but the optimal solution changes under language instructions. Through inheritance and continued optimization, our OmniFuse finally can be capable of defending against various composite degradation while supporting language-specified object highlighting.

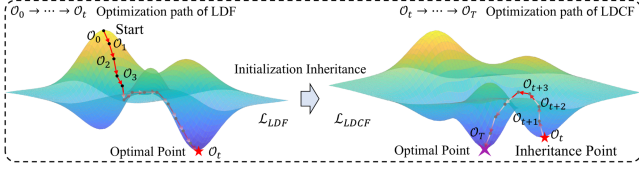


Fig. 6. Inherited optimization from LDF to LDCF.

IV. EXPERIMENTS AND RESULTS

In this section, we compare our OmniFuse with state-of-the-art image fusion methods across various challenging scenarios with composite degradation. We also evaluate the effectiveness of language control and its impact on both the visual and semantic aspects of the fused images. At the end, extensive ablation studies are conducted to analyze the role of the relevant technical designs.

A. Implementation Details

1) *Datasets*: To cover the various types of degradation, we construct the required training and testing dataset based on MFNet [28], FMB [30], and LLVIP [29]. Specifically, we introduce the concepts of a restoration pool and a degradation pool. The restoration pool consists of a series of image restoration algorithms (e.g., QuadPrior [38] and Diff-retinex [39]) that focus on constructing clean versions of available degraded multi-modal images. In contrast, the degradation pool includes various degradation functions (e.g., Gaussian noise, Poisson noise, and stripe), which aim to generate degraded versions of the available clean multi-modal images. Notably, each restored and degraded sample is visually inspected to select the best results from different algorithms, thereby not being limited by the performance of any specific algorithm. With the help of the restoration pool and a degradation pool, we construct a dataset consisting of 1500 clean-degraded multi-modal image pairs for training latent diffusion fusion. Notably, since the LLVIP dataset does not contain segmentation labels, we use 500 image pairs each from the MFNet and FMB datasets to sequentially train the subsequent language-driven controllable fusion. Corresponding testing covers four scenarios: cross-modal degradation scenario, infrared degradation scenario, visible degradation scenario, and non-degradation scenarios, utilizing a total of 170 pairs of images for evaluation.

2) *Training Details*: The autoencoder is first trained to identify an appropriate latent space. Then, the latent diffusion model is trained on clean-degraded paired data to embed the information restoration priori. Subsequently, we optimize LFFM to achieve latent diffusion fusion, enhancing the fusion model's robustness. Finally, inheriting the robustness of LDF, we continue training LFFM under different language instructions to achieve language-driven controllable fusion. At this stage, the process also involves iterative optimization with the semantic segmentation network. We use the AdamW optimizer with an initial learning rate of $1e^{-5}$ to update the parameters of all network modules. All experiments are conducted on four NVIDIA

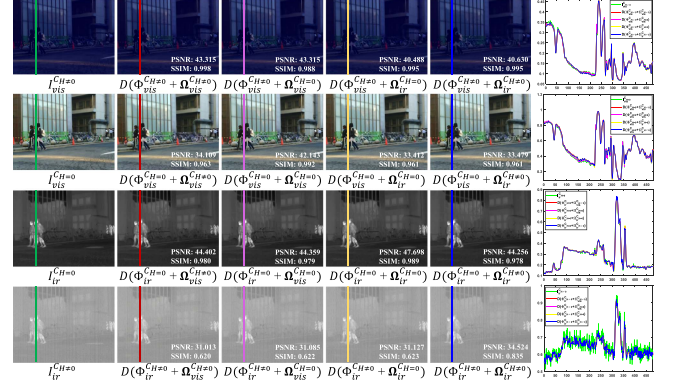


Fig. 7. Qualitative and quantitative verification of feature reorganization. Column 1: source images; Columns 2-5: reconstructed results under different reorganization configurations; Column 6: intensity scanning curves.

GeForce RTX 3090 GPUs with 24GB memory and one AMD EPYC 7H12 64-Core Processor CPU.

3) *Evaluation Strategy*: We employ seven state-of-the-art fusion methods for comparison, including LRRNet [40], Dif-fusion [16], DDFM [18], IF-FILM [23], Text-IF [7], MRFS [8], and our preliminary version DDBF [6]. These competitors encompass the latest technologies, e.g., diffusion models, and visual-language large models, representing the forefront of the image fusion field. Among them, Text-IF and DDBF support dynamic control of the fusion process. For Text-IF, we inform it of all the types of degradation present in the source images through textual input. For DDBF, we set a specific enhancement ratio based on the illumination condition of each instance to achieve better fusion performance. Besides, five objective metrics are used to evaluate fusion performance, including SD [41], PSNR, MI [42], SCD [43], and SSIM [44].

B. Feature Reorganization Verification

The autoencoder in our OmniFuse should meet two aspects of invariance: same-modal reorganization invariance and cross-modal reorganization invariance. To evaluate these functions, we conduct feature reorganization verification. Specifically, cross-modal degraded-clean image pairs $[I_{vis}^{C_H \neq 0}, I_{vis}^{C_H=0}, I_{ir}^{C_H \neq 0}, I_{ir}^{C_H=0}]$ are selected. We randomly exchange their latent features and multi-layer middle features, and then use the decoder for image reconstruction. Fig. 7 illustrates the reconstructed results for each type of reorganization. Firstly, whether for infrared images or visible images, feature reorganization within the same modality does not affect the reconstruction. In other words, whether the reconstructed image contains degradation depends entirely on whether the latent features originate from the degraded images. This proves that the latent features effectively represent the degradation attributes, laying a foundation for degradation removal in the latent space. Secondly, the multi-layer middle features exhibit good cross-modal invariance, as evidenced by the fact that cross-modal feature reorganization does not affect the reconstructed results. This allows us to use a simple fusion rule to aggregate them in the subsequent information fusion. To objectively assess the

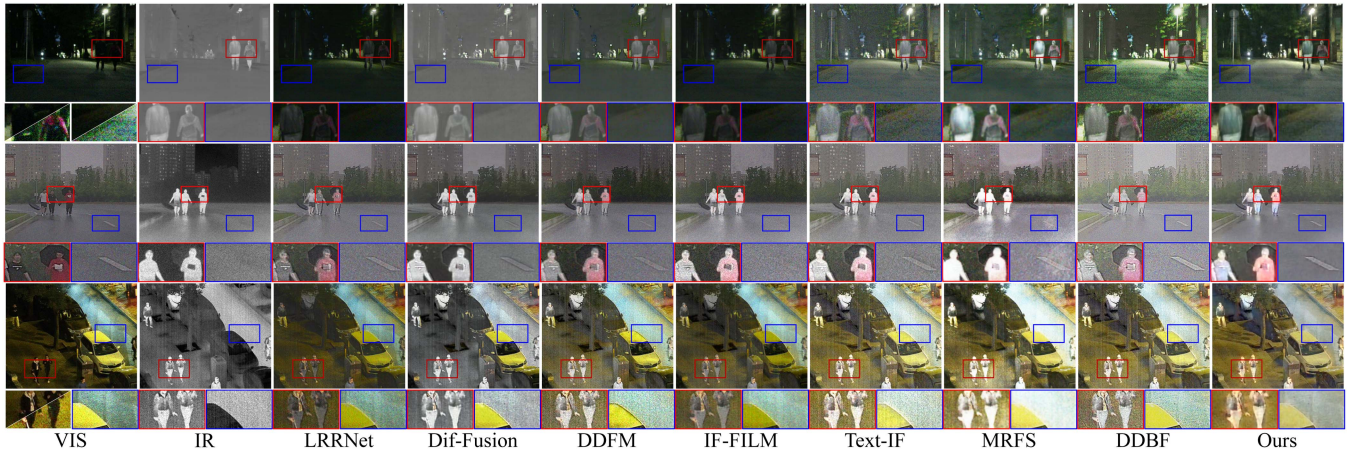


Fig. 8. Qualitative results in **cross-modal degradation** scenarios. In some extremely dark regions, we amplify the intensity of the source images to highlight the degradation hidden in the darkness.

TABLE III
QUANTITATIVE RESULTS IN **CROSS-MODAL DEGRADATION** SCENARIOS

Cross-Modal Degradation	MI $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SCD $\uparrow$	SD $\uparrow$
LRRNet (TPAMI'23)	1.370	0.544	62.507	0.981	6.175
Dif-Fusion (TIP'23)	2.137	0.657	63.112	0.844	6.327
DDFM (ICCV'23)	2.110	0.656	62.865	1.278	6.914
IF-FILM (ICML'24)	2.142	0.669	65.096	0.794	6.340
Text-IF (CVPR'24)	1.228	0.418	66.361	1.342	8.195
MRFS (CVPR'24)	2.094	0.712	64.749	1.329	7.921
DDBF (CVPR'24)	1.442	0.396	65.021	1.444	8.574
Ours	2.511	0.755	69.126	1.645	8.883

TABLE IV
QUANTITATIVE RESULTS IN **VISIBLE DEGRADATION** SCENARIOS

Visible Degradation	MI $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SCD $\uparrow$	SD $\uparrow$
LRRNet (TPAMI'23)	1.290	0.543	63.590	0.671	6.013
Dif-Fusion (TIP'23)	2.479	0.753	65.621	1.038	7.027
DDFM (ICCV'23)	2.286	0.659	64.435	1.107	7.393
IF-FILM (ICML'24)	1.758	0.469	63.698	1.012	7.369
Text-IF (CVPR'24)	1.652	0.422	65.379	1.222	9.221
MRFS (CVPR'24)	2.138	0.729	65.359	1.274	7.971
DDBF (CVPR'24)	2.286	0.415	63.231	1.173	8.914
Ours	2.537	0.768	66.193	1.360	8.788

reorganization invariance, we also provide intensity scanning curves, which indicate that the various reconstructed results closely align with the source images. Moreover, the PSNR and SSIM scores also confirm the excellent reconstruction capability of our autoencoder.

C. Cross-Modal Degradation Scenarios

Firstly, we conduct comparative experiments in cross-modal degradation scenarios, in which both infrared and visible images suffer from severe degradation and lose a significant amount of scene information. In this situation, recovering scene information and achieving high-quality fusion is highly challenging. Fig. 8 presents the qualitative comparison, in which the visible images experience low light, noise, and color cast, while the infrared images experience low contrast, noise, and stripes. LRRNet, Dif-Fusion, DDFM, and IF-FILM are unable to remove any type of degradation, and their fused results exhibit poor visual appearance. Although Text-IF, MRFS, and DDBF can enhance illumination, they still struggle with composite noise. In contrast, our method can filter out these complex negative factors, effectively restoring and integrating scene information. Besides, as shown in Table III, our method ranks first across all five quantitative metrics. Overall, these results demonstrate the robustness of our proposed OmniFuse under extreme degradation scenarios.

D. Single Degradation Scenarios

The single degradation scenarios include the visible degradation scenarios and the infrared ones, where only one modality contains composite degradation while the other is clean.

1) *Visible Degradation Scenarios*: The qualitative comparison in the visible degradation scenarios is presented in Fig. 9. The quality of the visible images is poor, while the infrared images are relatively clean. In this setting, the degradation in the visible images is still transmitted into the results of LRRNet, Dif-Fusion, DDFM, IF-FILM, Text-IF, and DDBF. Although MRFS can remove a small amount of degradation, e.g., noise, it exhibits drawbacks like texture loss and thermal radiation overflow. By comparison, our OmniFuse simultaneously enhances scene illumination, removes complex noise, and corrects color cast. Moreover, it effectively integrates multi-modal information, preserving fine texture details and significant thermal objects. The quantitative comparison in Table IV shows that our OmniFuse achieves the highest score in four of five objective metrics. Therefore, we can conclude that our method also demonstrates the best robustness in the visible degradation scenarios.

2) *Infrared Degradation Scenarios*: The qualitative comparison in the infrared degradation scenarios is shown in Fig. 10. The infrared images suffer from composite degradation, e.g., low contrast, noise, and stripes, while the visible images are relatively clean. Although there is a small amount of degradation remaining in the results of LRRNet, Dif-Fusion, DDFM,

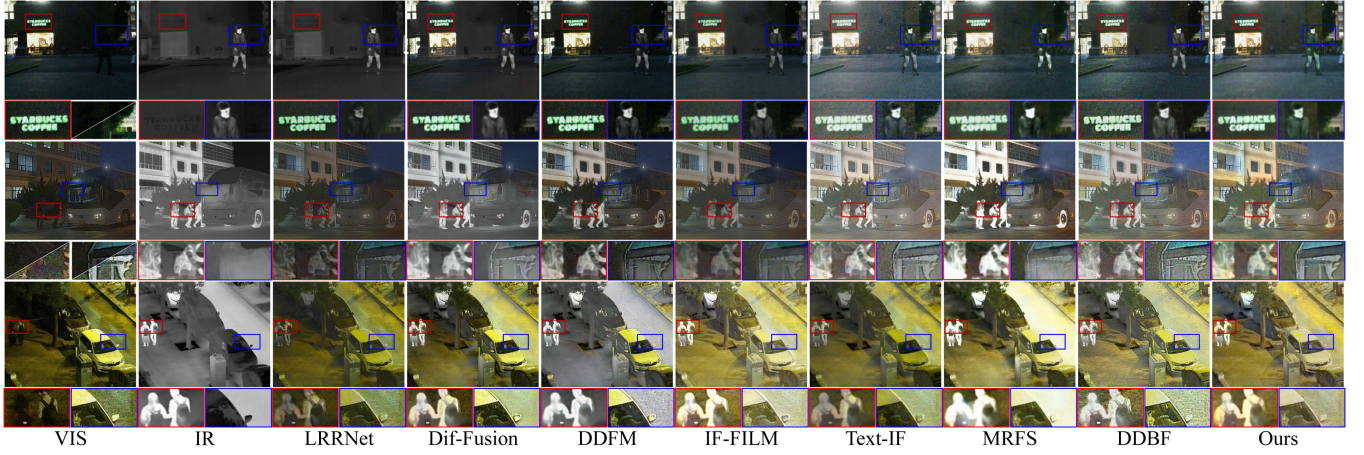


Fig. 9. Qualitative results in **visible degradation** scenarios. In some extremely dark regions, we amplify the intensity of the source images to highlight the degradation hidden in the darkness.

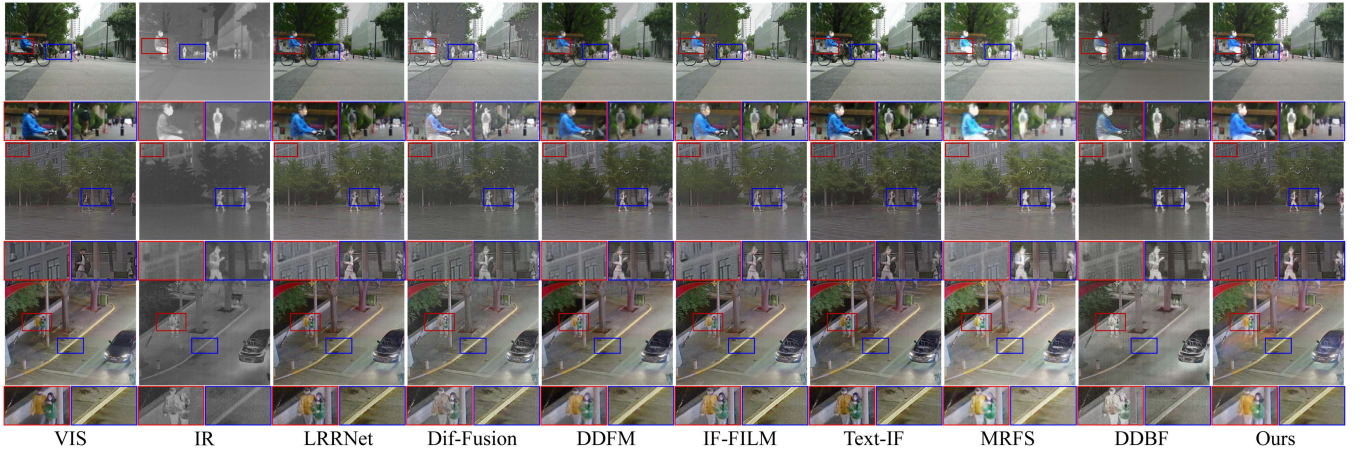


Fig. 10. Qualitative results in **infrared degradation** scenarios.

IF-FILM, and Text-IF, it has little impact on visual performance. This is because these methods primarily focus on preserving visible information, which helps in avoiding the transmission of degradation from the infrared modality. However, this still leads to their inability to maintain the saliency of thermal radiation. While MRFS can highlight thermal objects, it presents a severely blurred appearance. In addition, DDBF shows poor robustness, as both noise and stripes are still present in its fused results. In contrast, our OmniFuse can recover high-contrast thermal radiation information from degraded infrared images and excellently fuse it with visible information. The quantitative comparison in Table V shows that our method ranks first in three of the five metrics, further demonstrating its advantages.

E. Non-Degradation Scenarios

We further relax the difficulty to require that the source images contain as few degradation factors as possible and have a clean appearance. Under this condition, the comparison can focus on the ability of each method in information fusion. The qualitative

TABLE V
QUANTITATIVE RESULTS IN **INFRARED DEGRADATION** SCENARIOS

Infrared Degradation	MI $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SCD $\uparrow$	SD $\uparrow$
LRRNet (TPAMI'23)	2.181	0.566	65.482	1.132	9.095
Dif-Fusion (TIP'23)	2.415	0.567	65.761	1.299	9.569
DDFM (ICCV'23)	2.286	0.624	63.219	1.438	8.945
IF-FILM (ICML'24)	1.841	0.370	64.601	1.301	9.408
Text-IF (CVPR'24)	2.442	0.443	66.339	1.315	9.781
MRFS (CVPR'24)	2.749	0.623	61.785	1.318	10.027
DDBF (CVPR'24)	1.570	0.451	56.048	0.708	8.904
Ours	3.406	0.704	66.679	1.318	9.746

comparison in the non-degradation scenarios is presented in Fig. 11. Our OmniFuse has better information balance capability compared to other methods. First, LRRNet, DDFM, IF-FILM, and Text-IF perform well in preserving structural textures, but they reduce the saliency of thermal objects. Besides, Dif-Fusion, MRFS, and DDBF can effectively present thermal radiation information, but some background textures are weakened or even lost. In contrast, our method can combine the advantages of these competitors. It can derive a more natural fusion appearance, in

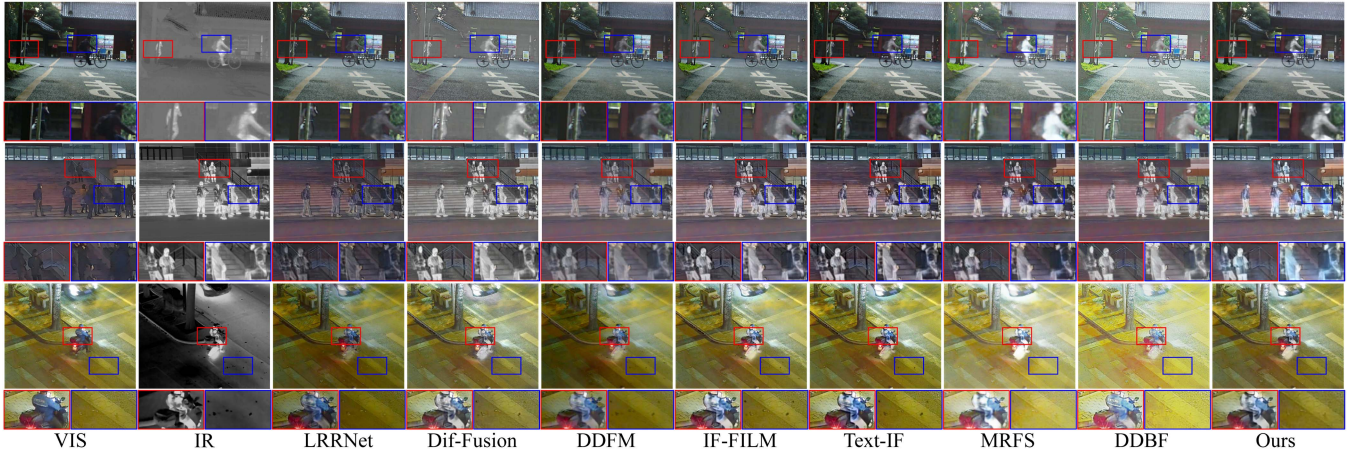
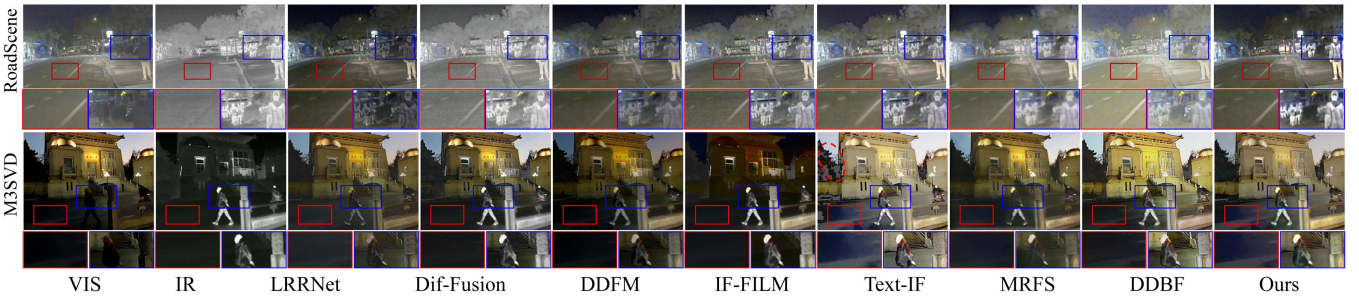
Fig. 11. Qualitative results in **non-degradation** scenarios.

Fig. 12. Qualitative generalization towards RoadScene [31] and M3SVD [45] datasets.

TABLE VI
QUANTITATIVE RESULTS IN **NON-DEGRADATION** SCENARIOS

Non-Degradation	MI $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SCD $\uparrow$	SD $\uparrow$
LRRNet (TPAMI'23)	3.279	0.785	64.774	1.314	8.972
Dif-Fusion (TIP'23)	2.818	0.764	64.101	1.262	8.452
DDFM (ICCV'23)	3.133	0.762	63.833	1.575	8.766
IF-FILM (ICML'24)	3.418	0.779	64.923	1.322	8.684
Text-IF (CVPR'24)	3.185	0.760	62.307	1.291	9.065
MRFS (CVPR'24)	3.078	0.758	62.075	1.262	9.146
DDBF (CVPR'24)	3.058	0.675	59.821	1.074	9.454
Ours	3.453	0.786	64.632	1.322	9.373

TABLE VII
QUANTITATIVE GENERALIZATION TOWARDS ROADSCENE [31] DATASET

Generalization	MI $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SCD $\uparrow$	SD $\uparrow$
LRRNet (TPAMI'23)	2.889	0.635	60.902	1.347	9.984
Dif-Fusion (TIP'23)	2.980	0.699	63.590	1.194	10.356
DDFM (ICCV'23)	2.912	0.697	60.232	1.525	10.149
IF-FILM (ICML'24)	3.169	0.696	62.696	1.212	10.399
Text-IF (CVPR'24)	3.151	0.711	63.768	1.437	10.475
MRFS (CVPR'24)	3.121	0.720	63.792	1.326	10.841
DDBF (CVPR'24)	2.848	0.661	62.495	1.120	10.370
Ours	3.416	0.724	64.020	1.199	10.409

which the sharpness of the background texture and the salience of the foreground objects are well-balanced. The quantitative comparison in Table VI shows that OmniFuse achieves two best scores and two second-best scores. These results demonstrate that, when solely comparing the capabilities in multi-modal information fusion, our method can still maintain a leading performance.

F. Generalization Evaluation

Next, we test the trained model on the RoadScene [31] and M3SVD [45] datasets to evaluate the generalization performance of these fusion methods.

The qualitative results on the RoadScene dataset are provided in the first row of Fig. 12. Our method shows a significant advantage in visual performance. Firstly, OmniFuse generates fused images with excellent contrast, significantly surpassing other comparative methods. This high contrast characteristic is evident not only in the thermal objects but also in the scene background. Secondly, OmniFuse also exhibits outstanding performance in structure preservation. The quantitative comparison in Table VII demonstrates that OmniFuse achieves three best scores of five metrics, reflecting its promising fusion performance.

We provide the qualitative results on the M3SVD dataset in the second row of Fig. 12. Our method still exhibits superior degradation resistance compared to other comparative methods, producing a more natural visual appearance. Specifically, in low-light scenarios, our OmniFuse achieves stronger information

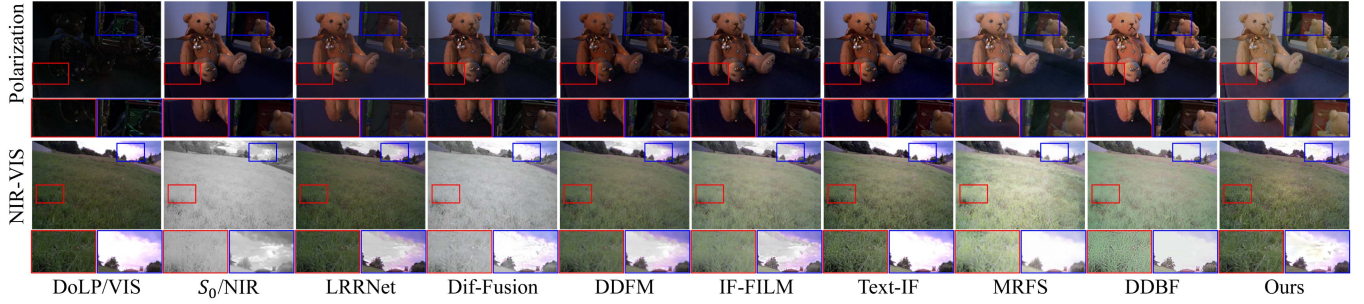


Fig. 13. Qualitative applications to the task of polarization image fusion and near-infrared and visible image fusion.

TABLE VIII
QUANTITATIVE GENERALIZATION TOWARDS M3SVD [45] DATASET

Generalization	MI ↑	SSIM ↑	PSNR ↑	SCD ↑	SD ↑
LRRNet (TPAMI'23)	4.088	0.665	64.720	1.636	10.011
Dif-Fusion (TIP'23)	4.019	0.630	64.430	1.763	10.678
DDFM (ICCV'23)	3.518	0.650	61.942	1.762	10.028
IF-FILM (ICML'24)	4.738	0.631	55.846	0.580	6.934
Text-IF (CVPR'24)	4.377	0.597	61.969	1.765	10.979
MRFS (CVPR'24)	3.736	0.650	64.845	1.762	10.461
DDBF (CVPR'24)	4.000	0.523	58.770	1.659	11.785
Ours	4.428	0.650	65.073	1.768	10.801

restoration ability, such as tree canopies and roads. The quantitative results in Table VIII further prove this point, in which OmniFuse also achieves two best scores and two second-best scores. Overall, these results prove that our method has good generalization ability.

G. Application to Other Fusion Tasks

The abilities of degradation removal and significant information integration in our OmniFuse can be understood as universal designs, which are widely needed by various multi-modal image fusion tasks. Therefore, our OmniFuse can theoretically be applied to other fusion tasks as well. To validate this, we test OmniFuse on the polarization image fusion and the near-infrared and visible image fusion.

1) *Polarization Image Fusion*: Polarized image fusion refers to the technology of fusing an intensity image S_0 and a degree of linear polarization (DoLP) image to enhance the perception of scene details. We perform comparative experiments on polarized image datasets [46], [47]. The quantitative results are presented in the first row of Fig. 13, where OmniFuse shows better visualization performance than others. First, the degradation removal capability of our method allows it to improve poor lighting conditions and correct color cast. Second, our OmniFuse effectively transmits structural information from the DoLP image to the fused image, resulting in a visually refined output. In contrast, other methods result in a certain degree of smoothing effect and color distortion. Moreover, in the quantitative comparison in Table IX, OmniFuse achieves the best scores in three out of five metrics. These results confirm that our OmniFuse can be effectively applied to address the task of polarized image fusion.

TABLE IX
QUANTITATIVE APPLICATIONS TO THE POLARIZATION IMAGE FUSION

Polarization	MI ↑	SSIM ↑	PSNR ↑	SCD ↑	SD ↑
LRRNet (TPAMI'23)	3.340	0.715	66.093	1.522	10.179
Dif-Fusion (TIP'23)	3.241	0.719	66.393	1.514	10.117
DDFM (ICCV'23)	2.895	0.719	65.031	1.777	10.108
IF-FILM (ICML'24)	3.628	0.729	65.756	1.701	10.274
Text-IF (CVPR'24)	3.381	0.744	66.840	1.579	10.044
MRFS (CVPR'24)	3.260	0.731	65.000	1.503	10.270
DDBF (CVPR'24)	3.551	0.636	60.221	1.473	10.566
Ours	3.892	0.744	67.121	1.578	10.147

TABLE X
QUANTITATIVE APPLICATIONS TO THE NIR-VIS IMAGE FUSION

NIR-VIS	MI ↑	SSIM ↑	PSNR ↑	SCD ↑	SD ↑
LRRNet (TPAMI'23)	4.283	0.770	63.672	1.203	10.103
Dif-Fusion (TIP'23)	3.955	0.810	65.451	1.061	10.196
DDFM (ICCV'23)	4.273	0.783	65.066	1.310	9.984
IF-FILM (ICML'24)	4.411	0.696	66.009	1.316	10.409
Text-IF (CVPR'24)	4.359	0.824	65.672	1.611	10.322
MRFS (CVPR'24)	3.775	0.788	63.792	1.157	10.873
DDBF (CVPR'24)	4.084	0.757	62.495	0.968	11.177
Ours	4.489	0.825	66.015	1.213	10.447

2) *Near-Infrared and Visible Image Fusion*: The goal of near-infrared (NIR) and visible image fusion is to use near-infrared information to assist in enhancing the quality of visible images. We apply OmniFuse to the near-infrared and visible image fusion datasets [48], and the qualitative results are shown in the second row of Fig. 13. First, OmniFuse is better at utilizing the texture details from near-infrared images to enhance visible images, such as clouds in the sky. Second, OmniFuse excels at maintaining color consistency between the fused image and the visible image, as seen in elements like grass. In comparison, other methods cause some details to be diminished or lost, and disrupt the original color distribution of the visible image. The quantitative results in Table X show that our method achieves the best scores in three out of five metrics. These results indicate that our method can also be applied to the task of near-infrared and visible image fusion with competitive performance.

H. Visual Gains of Language Control

In all the above experiments, our method defaults to empty language instructions. In this section, we validate the flexibility of our OmniFuse by highlighting semantic objects of interest

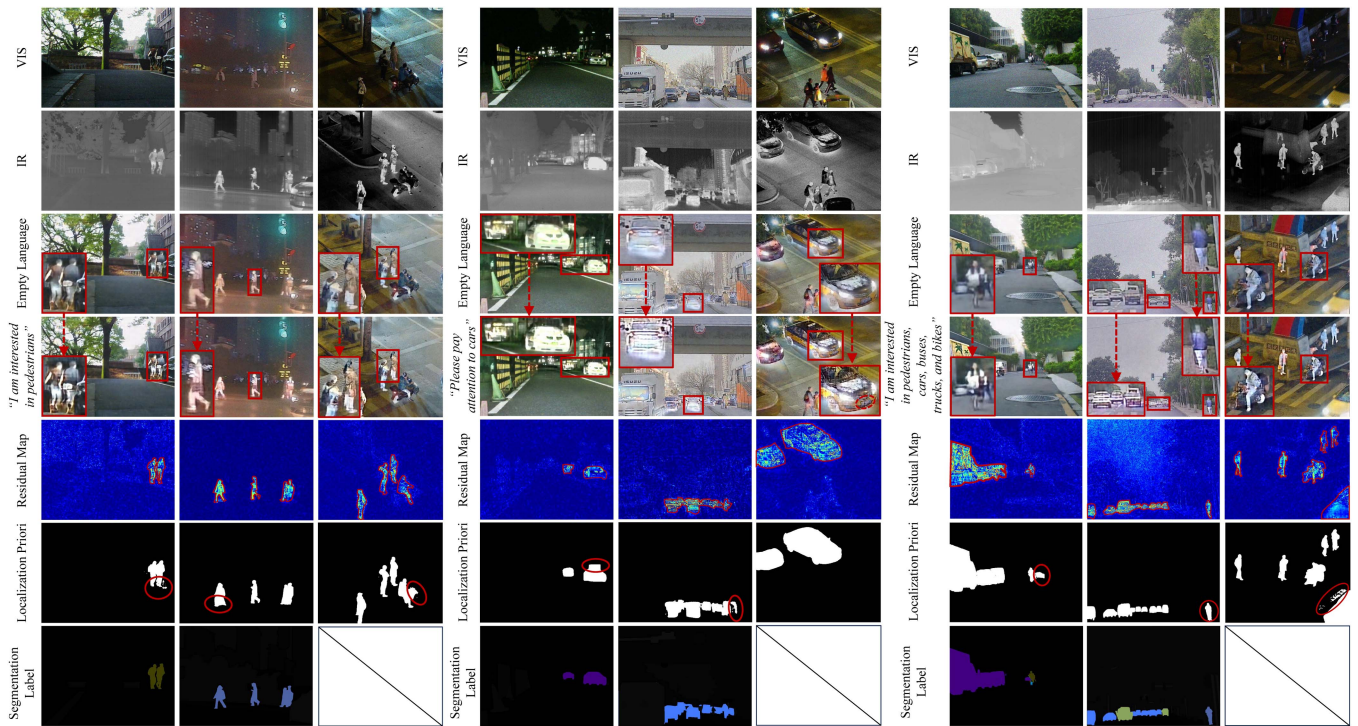


Fig. 14. Qualitative visual gains under different language instructions. From top to bottom: visible images, infrared images, fused images with no language guidance, fused images with specific language guidance, residual maps between the results with and without language instructions, location priori, and corresponding segmentation labels. Notably, since the LLVIP dataset lacks segmentation labels, it is shown as empty.

based on language instructions. Notably, to demonstrate that our method is robust and flexible at the same time, the visual gain of language control is evaluated in challenging cross-modal degradation scenarios. We input the language instructions: “*I am interested in pedestrians*”, “*Please pay attention to cars*”, and “*I am interested in pedestrians, cars, buses, and bikes*” into OmniFuse. The visual gains in their fused results, compared to the empty instruction, are shown in Fig. 14. As always, our method maintains the fusion robustness, effectively removing the degradation present in the source images. More importantly, driven by language instructions, the saliency of the specified object is effectively enhanced. First, the brightness of the object of interest is increased. For example, when our method is informed that pedestrians are of interest, the pixel intensity in regions occupied by pedestrians is noticeably amplified. Meanwhile, the sharpness of the object of interest is intensified. For instance, when we instruct OmniFuse to focus on cars, the license plate numbers in the sixth column even change from blurred to clear. Moreover, the results in columns seven to nine indicate that our method supports the simultaneous highlighting of multiple semantic categories, showcasing strong practicality.

Moreover, the residual maps show that pixel-level changes are concentrated in the regions specified by the language, providing strong evidence for the control accuracy of our method. In the regions of non-interest, the residual map is low intensity, indicating that our method inherits their robust fusion appearance. Besides, we visualize the location priori produced by Grounded-SAM along with the segmentation labels. The location priori is not perfect in certain cases, potentially encountering issues of

omissions, redundancies, or incomplete areas. At this point, OmniFuse demonstrates strong correction capability, ensuring that the controlled regions remain consistent with the segmentation labels.

The language control in our OmniFuse enables the information in the regions of interest within the fused images to exceed that in the source images. Therefore, the conventional metrics that consider the source images as the reference are no longer applicable in this case. Instead, we introduce the CLIP Score [10]. Specifically, we manually create captions describing the scene content for the test images based on visual perception. Then, we calculate the similarity scores between the captions and the fused images generated with various language instructions. Such a score can reflect whether the modulated fused images align with the human visual understanding of the scene, providing a new perspective for evaluating image fusion performance, especially for the popular generative fusion paradigms. The objective results in Fig. 15 demonstrate that our method can achieve a higher CLIP score as the categories of language-specified objects increase.

I. Semantic Gains of Language Control

OmniFuse can also provide semantic gains driven by language instructions. Such gains originate from low-level visual saliency improvement, along with high-level semantic injection driven by the segmentation task. To validate the semantic benefits of language control, we conduct the evaluation on two downstream tasks: object detection and semantic segmentation.

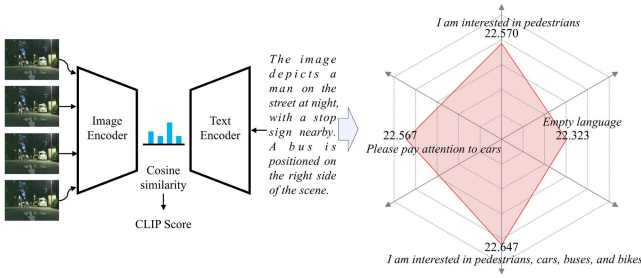


Fig. 15. Quantitative visual gains under different language instructions.

TABLE XI
QUANTITATIVE SEMANTIC GAINS UNDER LANGUAGE INSTRUCTIONS ON THE TASK OF OBJECT DETECTION

Detection	Precision	Recall	mAP@.5	mAP@.75	mAP@.5:.95
VIS	0.887	0.796	0.856	0.577	0.538
IR	0.931	0.896	0.919	0.684	0.611
LRRNet (TPAMI'23)	0.958	0.855	0.907	0.668	0.599
Dif-Fusion (TIP'23)	0.863	0.899	0.909	0.680	0.592
DDFM (ICCV'23)	0.949	0.897	0.915	0.729	0.628
IF-FILM (ICML'24)	0.967	0.895	0.927	0.728	0.629
Text-IF (CVPR'24)	0.949	0.894	0.931	0.733	0.631
MRFS (CVPR'24)	0.932	0.870	0.911	0.737	0.629
DDBF (CVPR'24)	0.953	0.897	0.922	0.730	0.626
Ours (Empty)	0.950	0.875	0.921	0.741	0.633
Ours(Prompt)	0.961	0.873	0.921	0.746	0.635

1) *Object Detection*: We first perform an evaluation on the object detection. Concretely, we use infrared images, visible images, and the fused images from each method to retrain YOLOv5 [49] for pedestrian detection. The experiments are conducted on the LLVIP dataset [29], with 1,000 images for training and 200 images for testing. Notably, our method includes two versions: one with empty language input and another with the prompt “I am interested in pedestrians”. The qualitative results in the first row of Fig. 16 demonstrate that OmniFuse achieves the best detection performance, accurately locating the various pedestrians while also having higher confidence. In contrast, other competitors exhibit missed detections. Besides, after receiving the language instruction, our method shows improvement in detection confidence for each pedestrian compared to the empty language version. The quantitative results in Table XI show that our OmniFuse achieves the best mAP@.5:.95 with the help of the language prompt. These results confirm that language control indeed enhances OmniFuse’s focus on semantic quality.

2) *Semantic Segmentation*: We further retrain SegNext [36] on infrared images, visible images, and the fused images from each method to evaluate their semantic quality. The training and testing data consist of 784 and 393 images from the MFNet dataset [28], respectively. Apart from providing OmniFuse with an empty language, we also feed it the prompt “I am interested in cars, people, bikes, ...” to emphasize the semantic categories in the MFNet dataset. The qualitative results are presented in the second row of Fig. 16, in which our method achieves segmentation regions with better completeness. Especially after the input of the prompt, the segmentation accuracy of our OmniFuse is further improved. The quantitative results in Table XII also display that the language prompt provides our method with the

TABLE XII
QUANTITATIVE SEMANTIC GAINS UNDER LANGUAGE INSTRUCTIONS ON THE TASK OF SEMANTIC SEGMENTATION

Segmentation	Car	Person	Bike	Curve	Car Stop	Guar.	Cone	Bump	mIoU
VIS	88.25	61.98	63.96	39.22	37.57	0.00	47.67	40.73	53.02
IR	86.58	71.36	56.32	41.26	25.53	9.06	42.78	51.15	53.54
LRRNet	88.14	72.29	65.00	42.70	28.69	5.86	50.94	51.50	55.92
Dif-Fusion	88.35	72.06	65.48	45.31	26.22	5.69	48.43	48.00	55.30
DDFM	88.72	71.88	64.89	44.46	27.16	0.00	56.08	47.98	55.48
IF-FILM	88.51	71.98	64.15	46.33	30.21	1.99	50.77	51.24	55.92
Text-IF	88.66	71.06	63.94	45.70	28.57	4.32	54.65	48.86	55.98
MRFS	88.47	70.91	64.16	42.38	33.58	0.02	52.25	39.72	54.40
DDBF	88.92	70.30	63.38	46.42	29.60	7.06	53.66	43.38	55.64
Ours (Empty)	87.77	68.46	62.60	37.19	34.80	11.09	55.33	48.96	56.02
Ours (Prompt)	87.27	69.73	63.87	37.72	33.69	13.51	53.81	48.46	56.23

highest average scores. Notably, an interesting phenomenon is that the IOU of the “car” category shows a slight decrease after the prompt is entered. We attribute this to the ambiguity in the naming of some categories in MFNet, e.g., “Curve” and “Bump”. The inaccurate emphasis on these ambiguously named categories by our method may cause some level of interference with others. Nevertheless, the language prompt has indeed led to an increase in the average segmentation accuracy for OmniFuse.

J. Ablation Studies

The main contributions of our OmniFuse are twofold: the latent diffusion fusion and the language-driven controllable fusion. Next, we conduct ablation studies on specific designs within them to clarify their roles.

1) *Latent Diffusion Fusion*: The ablation studies on LDF yield five model variants. **Model I**: replaces the significant contrast loss $\mathcal{L}_{SC}$ with a proportional intensity loss [50]; **Model II**: replaces the sharpened texture loss $\mathcal{L}_{ST}$ with a proportional gradient loss [50]; **Model III**: replaces the hue fidelity loss $\mathcal{L}_{HF}$ with YCbCr strategy [8]; **Model IV**: specify the fused multi-layer middle features in Eq. (13) as $\Omega_f = \Omega_{vis}$; **Model V**: $\Omega_f = \Omega_{ir}$; **Model VI**: $\Omega_f = 0.3 \times \Omega_{vis} + 0.7 \times \Omega_{ir}$; **Model VII**: $\Omega_f = 0.7 \times \Omega_{vis} + 0.3 \times \Omega_{ir}$; **Model VIII**: $\Omega_f = \max(\Omega_{vis}, \Omega_{ir})$; **Model IX**: $\Omega_f = \text{var}(\Omega_{vis}, \Omega_{ir})$ following the MaxFusion [51]; **Model X**: uses a direct generation strategy instead of a weighting strategy in Eq. (12); **Model XI**: removes skip connections between the encoder and decoder. The qualitative results of these ablation studies are presented in Fig. 17. Firstly, replacing $\mathcal{L}_{SC}$ and $\mathcal{L}_{SC}$ results in a reduction in the saliency of thermal objects and a smoothing of structures, respectively. Secondly, removing $\mathcal{L}_{HF}$ causes color distortion in the fused image, especially in regions with thermal radiation. The quantitative results in Table XIII also confirm that removing them causes score reduction. Thirdly, using linear weighting (Models IV-VII and the Full Model) to aggregate multi-level middle features does not cause significantly different fusion appearances, while employing nonlinear aggregation strategies (Models VIII and IX) leads to negative effects. Besides, when the direct generation strategy is used to fuse latent features or remove skip connections, the visual fusion performance is degraded. The quantitative results in Table XIII further prove the above conclusions.

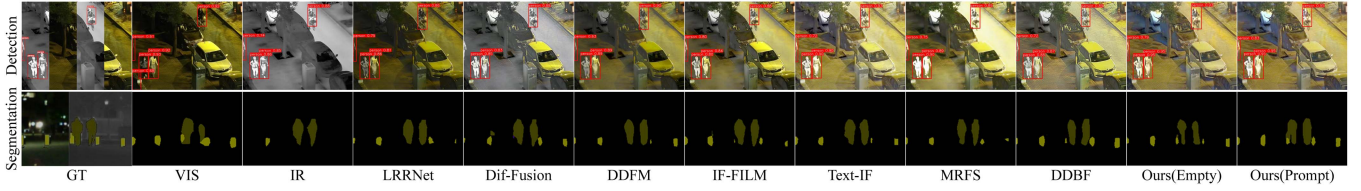


Fig. 16. Qualitative semantic gains under language instructions on the tasks of object detection and semantic segmentation.

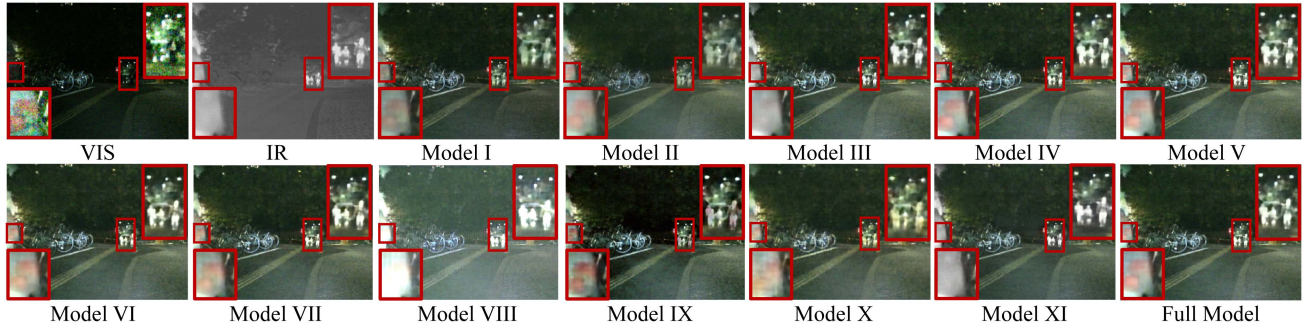


Fig. 17. Qualitative results of ablation studies on the latent diffusion fusion.

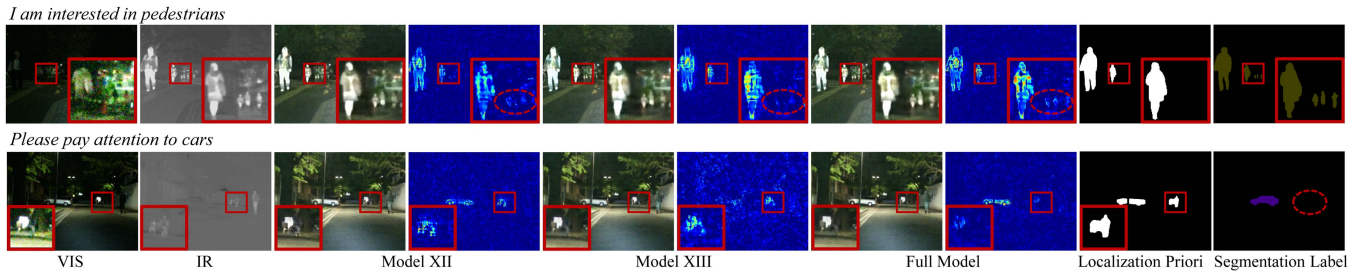


Fig. 18. Qualitative results of ablation studies on the language-driven controllable fusion.

TABLE XIII
QUANTITATIVE ABLATION STUDIES ON LATENT DIFFUSION FUSION

Ablation on LDF	MI $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SCD $\uparrow$	SD $\uparrow$
Model I	2.036	0.768	67.004	1.543	7.969
Model II	2.342	0.783	68.804	1.585	8.856
Model III	2.490	0.756	69.047	1.643	8.867
Model IV	2.480	0.747	69.065	1.645	8.881
Model V	2.465	0.752	69.009	1.638	8.871
Model VI	2.497	0.756	69.123	1.641	8.870
Model VII	2.508	0.755	69.137	1.644	8.863
Model VIII	2.153	0.603	60.701	1.594	8.736
Model IX	2.085	0.583	65.917	1.585	8.726
Model X	2.266	0.755	69.161	1.624	8.814
Model XI	2.112	0.746	67.238	1.395	8.132
Full Model	2.511	0.755	69.126	1.645	8.883

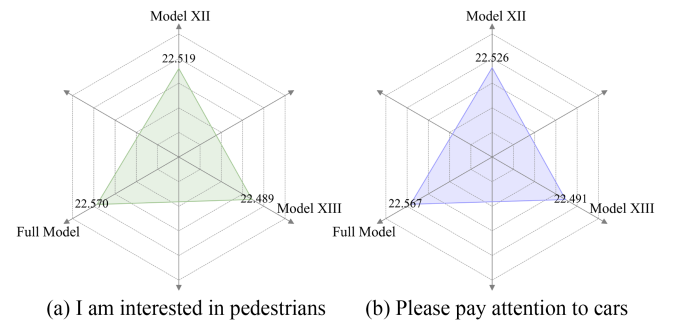


Fig. 19. Quantitative ablation studies on language-driven controllable fusion.

2) *Language-Driven Controllable Fusion*: The ablation studies of language-driven controllable fusion involve two main variants. **Model XII**: use the localization priori provided by Ground-SAM to construct the visual reference in Eq. (22) instead of using segmentation labels; **Model XIII**: remove the collaborative optimization between segmentation and fusion shown in Fig. 5. We implement these ablation studies under two

types of language instructions: “*I am interested in pedestrians*”, and “*Please pay attention to cars*”. The qualitative are shown in Fig. 18. First, when using the localization priori to construct visual references, the fusion model essentially learns to highlight the regions that are entirely consistent with the given conditions. At this point, the ability to correct imperfect localization priori is lost. Second, when the cooperative optimization is removed, the

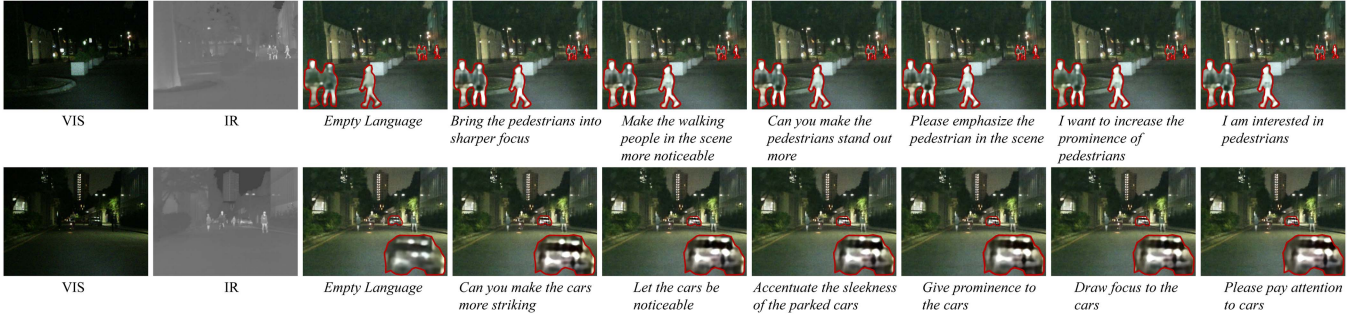


Fig. 20. Qualitative results under language instructions with different sentence structures.

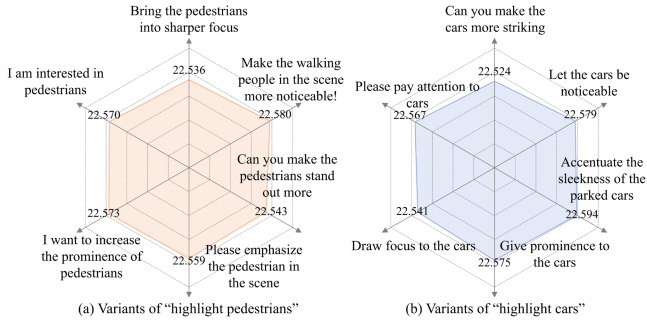


Fig. 21. Quantitative results under language instructions with different sentence structures.

fusion model struggles to accurately locate the objects of interest, resulting in some disorganization in the residual map. In contrast, the full model of our OmniFuse accurately identifies and highlights the specified areas even when the localization priori contains omissions or redundancies. Moreover, the quantitative results in Fig. 19 also indicate that the full model achieved the highest CLIP score. These results demonstrate the importance of these designs for language-driven controllable fusion.

3) *Instructions With Different Sentence Structures*: To verify the adaptability of our OmniFuse to different sentence structures, we perform ablation studies on language instructions. Concretely, we use six different prompts to highlight pedestrians and cars, and the qualitative results are shown in Fig. 20. It can be observed that even with variations in sentence structure, our OmniFuse is still able to accurately identify the objects of interest and enhance their saliency and texture. The quantitative results in Fig. 21 demonstrate that prompts with different expression forms yield only minor differences in CLIP scores. Overall, these results indicate that our method supports user inputs with different language instructions, as long as the underlying meaning remains consistent.

K. Efficiency Analysis

Efficiency is also an important consideration in evaluating the performance of fusion methods. Therefore, we calculate the average running time of each fusion method on image pairs with the size of 480×640 , as shown in Table XIV. Our OmniFuse

TABLE XIV
THE AVERAGE RUNNING TIME OF EACH FUSION METHOD ON THE IMAGE PAIRS WITH THE SIZE OF 480×640

Method	LRRNet	Dif-Fusion	DDFM	IF-FILM	Text-IF	MRFS	DDBF	Ours
Time/s	0.172	1.628	96.630	0.202	0.423	0.256	0.294	5.872

does not achieve the fastest speed, taking approximately 5.80 seconds to fuse each pair of multi-modal images. This may be due to its relatively complex internal architecture, which encompasses the autoencoder, diffusion model, language model, etc., resulting in a substantial computational load. Nevertheless, our method is still much faster than DDFM, which is also based on the diffusion model. Considering its excellent robustness and flexibility, OmniFuse remains highly competitive.

V. CONCLUSION AND DISCUSSION

This paper proposes a composite degradation-robust image fusion framework with language-driven semantics, aiming to overcome the robustness barrier and flexibility limitation in image fusion. Firstly, to enhance robustness, we develop a latent diffusion-based fusion paradigm that embeds the fusion function into the latent space of LDM, enabling seamless integration of the diffusion and fusion processes. This allows OmniFuse to effectively handle various composite degradation while ensuring high-quality fusion. Secondly, to improve flexibility, we design a language-driven controllable fusion strategy. By incorporating language-specified localization priori and customized visual-semantic constraints, the fusion module dynamically aggregates multi-modal features, enabling OmniFuse to flexibly highlight objects of interest based on language instructions. Extensive experiments demonstrate that our OmniFuse achieves attractive fusion performance with strong robustness and high flexibility.

A limitation of our method is that there is still room for improvement in its efficiency. In response to this challenge, we will advance the research of real-time image fusion in the future while maintaining the existing advantages of robustness and flexibility. One potential solution is to leverage the cross-modal information fusion priori to accelerate the sampling process of the diffusion model, thereby reducing the estimation difficulty of the composite degradation distribution.

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